

Crypto challenge - Injection

deuterium — 2021-07-25

Canonical HTML: <https://deut-erium.github.io/2021/07/25/injection.html>

Abstract

Crypto challenge - Injection, published in deuterium's blog.

Bijections are fun to look at and amusing especially to cryptographers, can you spot out one here?

```
from secret import flag

def nk2n(nk):
    l = len(nk)
    if l==1:
        return nk[0]
    elif l==2:
        i,j = nk
        return ((i+j)*(i+j+1))//2 + j
    return nk2n([nk2n(nk[:l-l//2]), nk2n(nk[l-l//2:])])

print(nk2n(flag))

# output
# 1066464516621568650416778516260128065562999836454777449496486613730252783905\
# 58656231590803591166516524516182041583860744711996793449978222571578932566\
# 51539240517205572748689616288529831032342817805470118893063573639935906790\
# 19094216260987077393364474718427466510193852
```

Enter the flag

The local SHA-256 checker requires JavaScript. The challenge itself does not.

| Interactive controls are omitted from this PDF. See the canonical HTML version.